Sample Project Documentation Template

Date	15 March 2026
Team ID	Batch 2023-27 | ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	5 Marks

1. Project Overview

Attribute	**Details**
Project Title	Rising Waters – ML-Based Flood Prediction System
Team ID	Batch 2023-27 | ACOE
Team Members	Venkatesh Balireddy | Archana Dhanani | Shivatmika Gandikota | Gode Siva Ramakrishna Durgaprasad | Bavisetty Gopi Krishna
Technology Track	AI / Machine Learning

2. Problem Statement

Floods are among the most devastating natural disasters globally, claiming thousands of lives and displacing millions every year. Conventional forecasting methods lack the predictive intelligence required for timely early warnings. This project builds a machine learning-powered flood prediction web application to provide accurate, real-time risk assessment.

3. Dataset Details

Attribute	**Details**
Dataset Size	5,000 records
Input Features	Temp, Humidity, Cloud Cover, ANNUAL, Jan-Feb, Mar-May, Jun-Sep, Oct-Dec, avgjune, sub

Attribute	Details
Target Variable	flood_occurred (0 = No Flood, 1 = Flood)
Source	Synthetic dataset based on real meteorological patterns
Split	80% Training (4,000 rows) | 20% Testing (1,000 rows)

4. Machine Learning Pipeline

Step	Activity	Output
1	Environment Setup	Python 3.11, all dependencies installed
2	Dataset Generation	flood_dataset.csv
3	Exploratory Data Analysis	4 visualisation charts
4	Preprocessing	Imputed nulls, scaled with StandardScaler
5	4-Algorithm Training	DT 78.17%, KNN 80.67%, RF 82.00%, Random Forest 95.65%
6	Best Model Selection	floods.save (Random Forest) + scaler.save
7	Deployment	Live web application on IBM Cloud

5. Web Application Pages

Page	Route	Description
Home Dashboard	/	Live prediction count, model accuracy comparison
Prediction Form	/predict	6-field form for meteorological inputs
Flood Chance Result	/result/flood	Red alert banner, 6 emergency action cards
No Flood Result	/result/noflood	Green safe banner, monitoring recommendations
Prediction History	/history	Full timestamped audit log
About	/about	5-layer system architecture, model info, team details

6. System Architecture Summary

Layer	Technology
User Layer	Web Browser
Presentation Layer	Jinja2 HTML Templates
Application Layer	Python 3.11 + Flask
ML Layer	Random Forest + Scikit-learn + Joblib
Data Layer	Pandas CSV
Deployment Layer	Docker + IBM Cloud Foundry

7. Team Contributions

S.No	Team Member	Role	Primary Contribution
1	Venkatesh Balireddy	Team Lead	Project lead, Flask app, IBM Cloud deployment
2	Archana Dhanani	Member	EDA, data visualisation, input validation utils
3	Shivatmika Gandikota	Member	Data preprocessing, StandardScaler pipeline
4	Gode Siva Ramakrishna Durgaprasad	Member	ML model training, Random Forest tuning, evaluation
5	Bavisetty Gopi Krishna	Member	HTML templates, CSS UI design